

DAVID ADAMS

Senior AI/Full Stack Engineer

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SUMMARY

Senior AI / Full-Stack Software Engineer with 10+ years of experience designing and delivering enterprise software platforms, cloud-native applications, AI-enabled systems, and data-intensive solutions. Specializes in production GenAI and LLM applications, Retrieval-Augmented Generation (RAG), distributed backend services, full-stack engineering, machine learning platforms, and scalable cloud architecture across AWS and Azure. Brings end-to-end engineering ownership across architecture, data flows, application development, deployment, observability, and production operations, with experience spanning public-sector AI, enterprise scientific research, and clinical data platforms.

EXPERIENCE

Senior Software Engineer

New York, NY

New York City Office of Technology & Innovation (NYC OTI)

06/2023 - Present

- Architected and delivered the MyCity AI chatbot platform with Retrieval-Augmented Generation (RAG), GPT-4, Azure OpenAI, Azure AI Search, and vector embedding pipelines, grounding generated responses in authoritative government content and establishing a production-ready foundation for citizen-facing information retrieval.
- Engineered LLM application workflows for conversational interactions, prompt orchestration, citation traceability, response validation, and responsible AI controls, strengthening answer transparency and establishing repeatable governance patterns for production generative-AI behavior.
- Designed distributed backend services with Python and FastAPI around persistent application state, asynchronous processing, document retrieval, and knowledge relationships, using PostgreSQL, Redis, Kafka, and Neo4j to support independently scalable AI workloads and dependable data flows.
- Built responsive AI product experiences with React and TypeScript, connecting conversational interfaces and application workflows to backend retrieval services while emphasizing accessibility, maintainability, and reliable performance for public digital services.
- Established cloud-native deployment practices with Azure, Docker, Kubernetes, AKS, and Terraform, improving release repeatability, infrastructure readiness, and operational consistency across distributed AI workloads.
- Implemented production observability with Grafana and Prometheus, extending service monitoring and operational visibility across AI components to support faster diagnosis, proactive reliability management, and sustained production readiness.
- Optimized AI service request handling and retrieval performance through API tuning and evaluation workflows, using production feedback to improve response behavior, system resilience, and readiness for sustained public usage.
- Mentored engineers through architecture reviews, design discussions, code reviews, and technical knowledge sharing, promoting engineering best practices and helping teams deliver maintainable, reliable AI platform capabilities in a production environment.

Senior Software Engineer

East Hanover, NJ/Remote

Novartis

06/2021 - 05/2023

- Architected enterprise AI solutions for scientific research workflows, applying machine learning engineering principles and model lifecycle practices to create scalable platform capabilities aligned with evolving research and analytics requirements.
- Built full-stack analytical applications with React, TypeScript, and Java services, enabling scientific and business teams to explore research results, experiment outcomes, and AI-driven insights through structured, interactive workflows.
- Engineered distributed data pipelines with Databricks, Apache Spark, and PySpark to transform complex scientific datasets into reusable analytical assets, improving the reliability, repeatability, and downstream usability of research data.
- Developed machine learning delivery pipelines using Azure Machine Learning, SageMaker, and MLflow, standardizing experimentation, model deployment, and lifecycle practices to improve consistency across production analytics workflows.
- Strengthened enterprise AI platform reliability through cloud architecture, security controls, monitoring, and production support practices across distributed computing environments, improving operational readiness for business-critical analytical workloads.
- Partnered with engineering, data science, and business stakeholders to translate complex research requirements into maintainable technical services, aligning platform architecture and delivery decisions with enterprise objectives.

EXPERIENCE

Software Engineer eClinical Solutions

Mansfield, MA/Remote

08/2015 - 05/2021

- Developed enterprise clinical data platforms with Java, Python, REST APIs, SQL, and PostgreSQL to support trial-data ingestion, validation, transformation, and workflow processing across complex healthcare application environments.
- Built web-based clinical applications with Angular, JavaScript, and TypeScript, giving researchers and operational teams structured interfaces for study workflow management, data review, and access to critical healthcare information.
- Created resilient integration services with Apache Kafka, Spark, and PySpark to process high-volume clinical datasets, improve interoperability, and support dependable data exchange between distributed enterprise systems.
- Delivered analytics and reporting workflows with Python-based data processing, transforming clinical and operational information into structured insights that supported research activities and recurring business analysis.
- Supported production healthcare platforms with AWS, Docker, Jenkins, automated deployment workflows, and monitoring practices, emphasizing application availability, controlled releases, operational stability, and dependable day-to-day delivery.
- Collaborated with cross-functional healthcare teams to translate clinical requirements into scalable software solutions while maintaining data accuracy, compliance readiness, and operational stability across evolving clinical workflows.

EDUCATION

B. S in Computer Science Trinity University

San Antonio, TX

09/2009 - 05/2013

M. S in Computer Science Trinity University

San Antonio, TX

09/2013 - 05/2015

SKILLS

AI/GenAI:

GPT, Azure OpenAI, RAG, vector search, embeddings, prompt engineering, guardrails, LangChain, MLflow, Azure Machine Learning, SageMaker, PyTorch, TensorFlow, scikit-learn

Full Stack Engineering:

Python, FastAPI, Java, REST APIs, React, Next.js, Angular, TypeScript, JavaScript, Node.js, Express, HTML5, CSS3, Material UI, React Router, Axios, Plotly, Power BI

Data / Distributed Systems:

SQL, PostgreSQL, Redis, MongoDB, Elasticsearch, Neo4j, Snowflake, Azure AI Search, Databricks, Apache Spark, PySpark, Amazon RDS, Amazon S3, AWS Glue, Amazon EMR, Pandas, NumPy, Apache Kafka

Cloud / Platform:

AWS (EC2, S3, Lambda, ECS, EKS, RDS, SageMaker, CloudWatch, IAM, CloudTrail, Glue, EMR), Azure (Azure ML, AKS, Azure DevOps, Azure Monitor), GCP (Vertex AI)

DevOps / Observability: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Grafana, Prometheus, Linux

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate (AI-102)